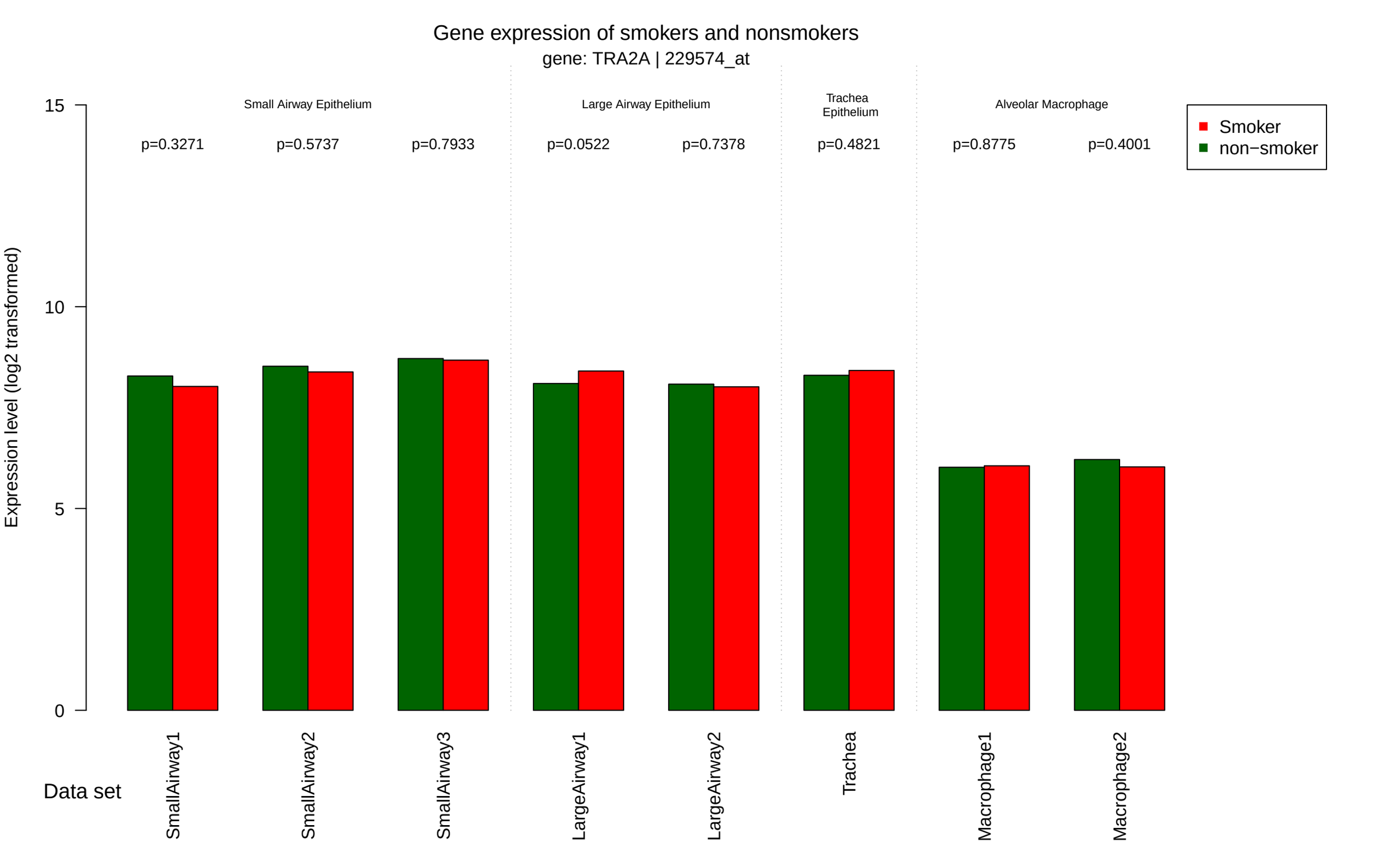
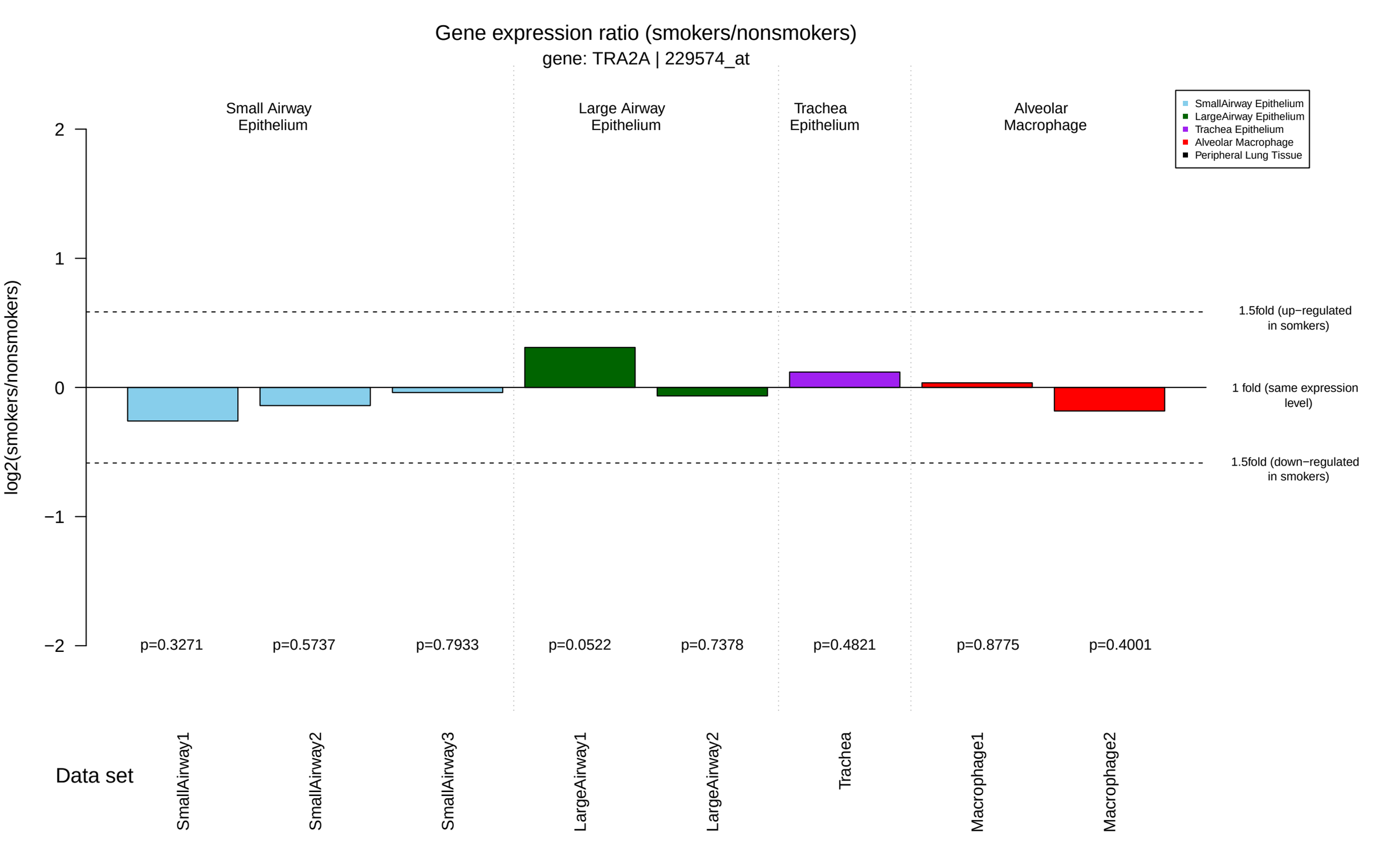
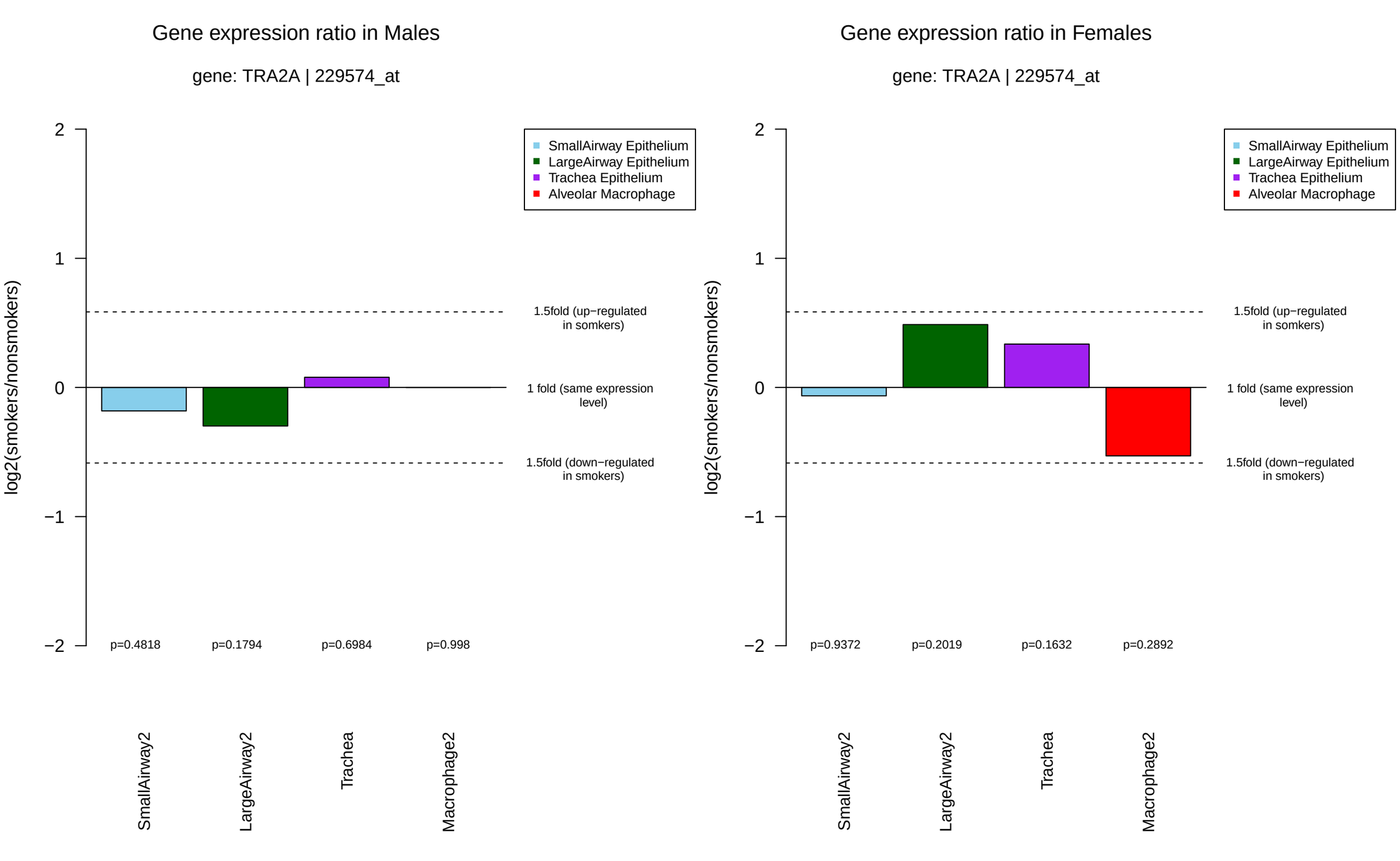
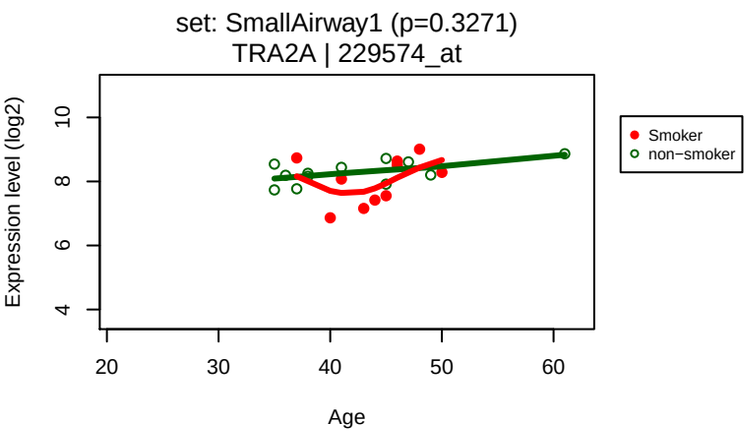
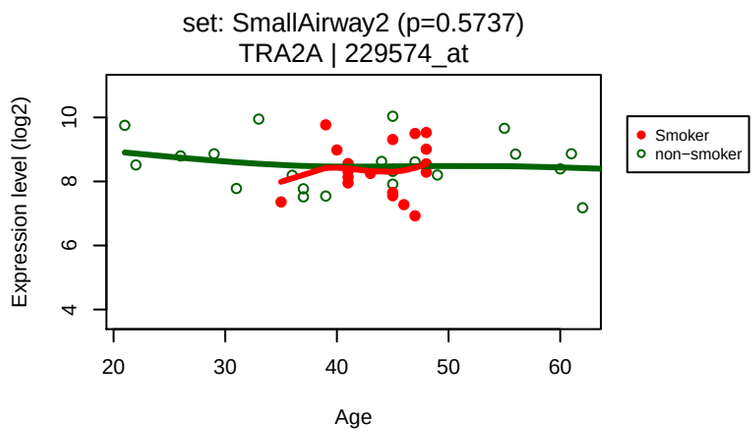
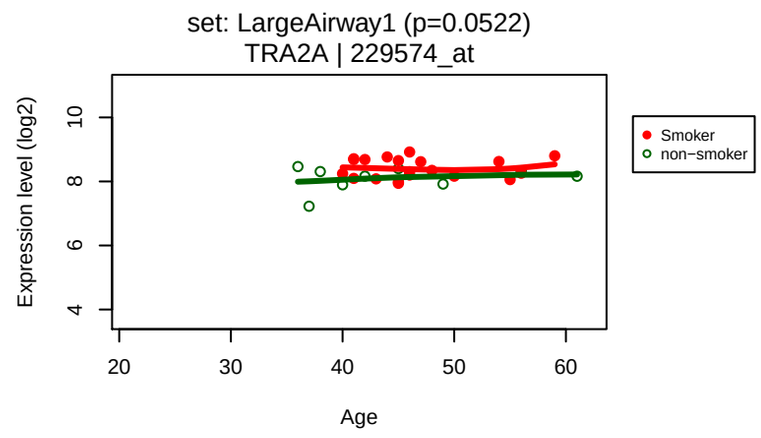

Gene expression of smokers and nonsmokers
gene: TRA2A | 229574_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3271
p=0.5737
p=0.7933
p=0.0522
p=0.7378
p=0.4821
p=0.8775
p=0.4001
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

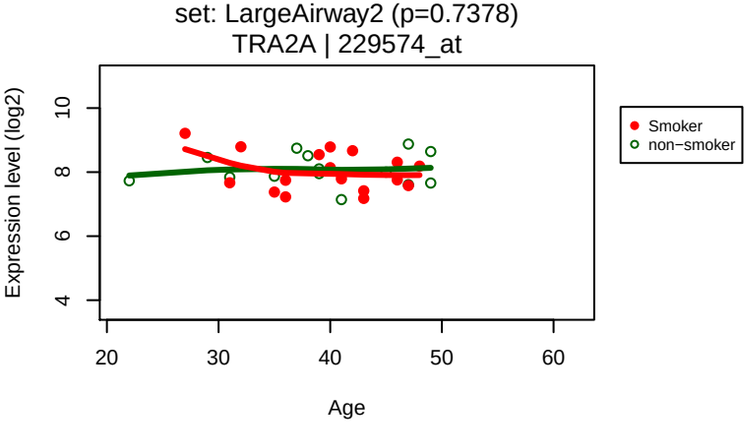

Gene expression ratio (smokers/nonsmokers)
gene: TRA2A | 229574_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3271
p=0.5737
p=0.7933
p=0.0522
p=0.7378
p=0.4821
p=0.8775
p=0.4001
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

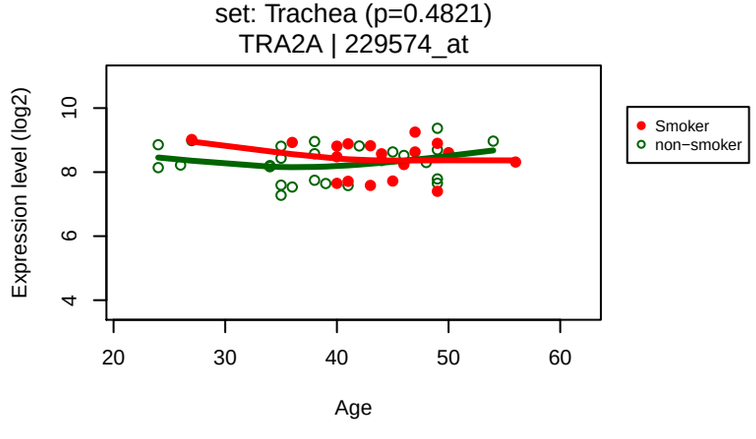

Gene expression ratio in Males
gene: TRA2A | 229574_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4818
p=0.1794
p=0.6984
p=0.998
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: TRA2A | 229574_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9372
p=0.2019
p=0.1632
p=0.2892
SmallAirway2
LargeAirway2
Trachea
Macrophage2

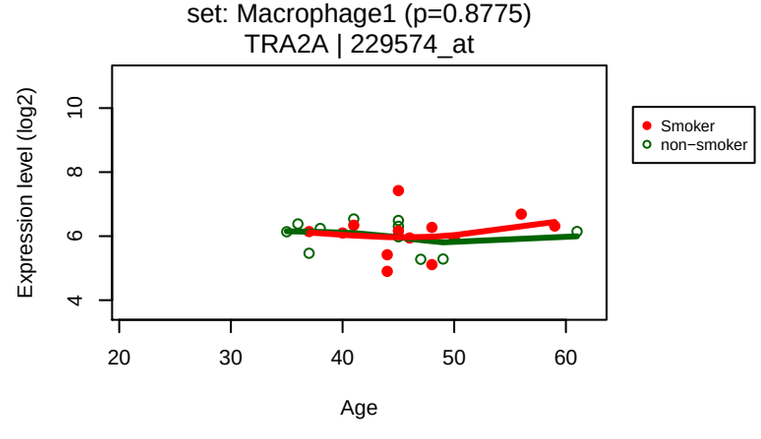

set: SmallAirway1 (p=0.3271)
TRA2A | 229574_at
Expression level (log2)
Age
Smoker
non−smoker

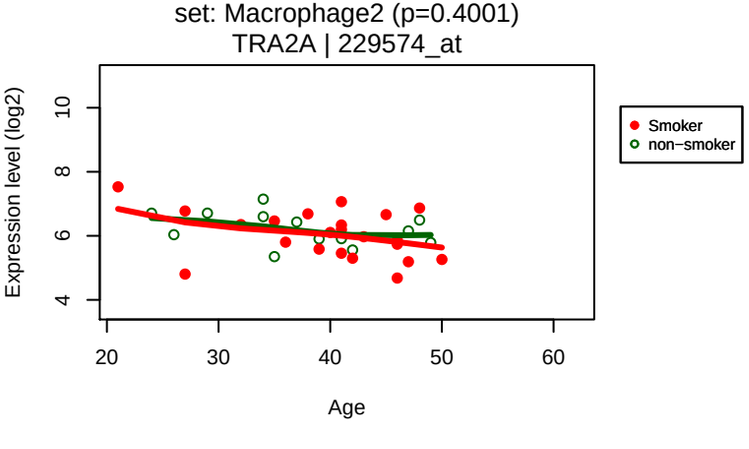

set: SmallAirway2 (p=0.5737)
TRA2A | 229574_at
Expression level (log2)
Age
Smoker
non−smoker

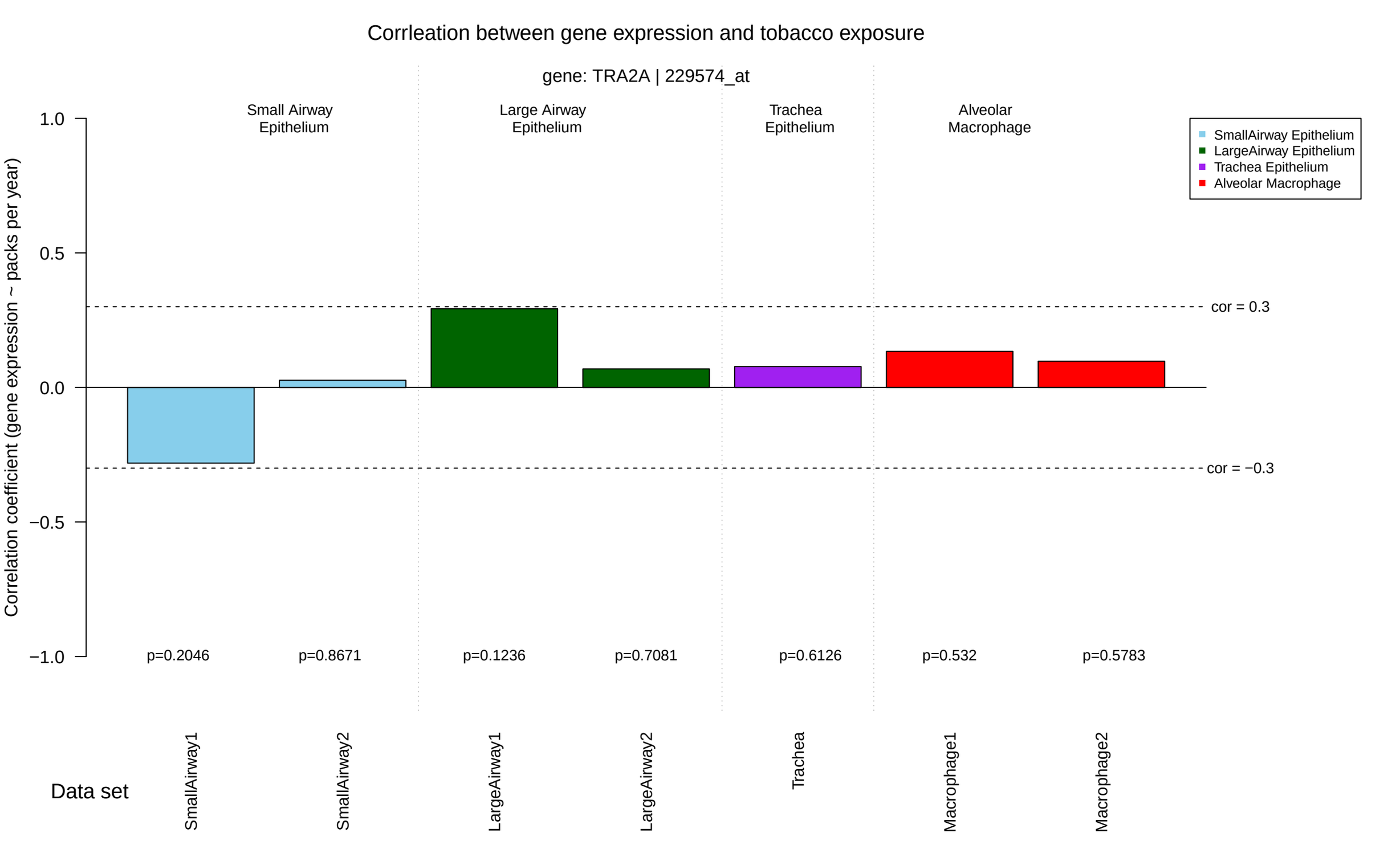

set: LargeAirway1 (p=0.0522)
TRA2A | 229574_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.7378)
TRA2A | 229574_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.4821)
TRA2A | 229574_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8775)
TRA2A | 229574_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4001)
TRA2A | 229574_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: TRA2A | 229574_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2046
p=0.8671
p=0.1236
p=0.7081
p=0.6126
p=0.532
p=0.5783
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2